



NORTH-HOLLAND

International Review of Economics and Finance
11 (2002) 331–333

IRF International
Review of
Economics
& Finance

Book review

Option Valuation Under Stochastic Volatility with Mathematica Code

Alan L. Lewis (2000). Newport Beach, California: Finance Press. ISBN: 0-9676732-0-1

Black Scholes options pricing formula assumes a constant volatility for the underlying asset. Applying modern statistical tests, most financial and commodity prices are found to exhibit time-varying volatility. To better describe the behavior of a security price in a continuous-time framework, two diffusion equations are needed, one for the security price itself, and the other for the evolution of the volatility. That is, a stochastic volatility process is preferred. This book considers the valuations of options under a time-homogenous stochastic volatility process. In Chapter 1, the author lays down the basic framework. Upon adopting an equity risk premium and a hedging risk premium (caused by the stochastic volatility) for the Girsanov transform, an option PDE in two space variables is derived.

To solve this PDE, the author proposes a “fundamental transform” method in Chapter 2. This method reduces the two space variables into one while adding a new transform variable. The new variable serves the classification purposes. It separates the solution into two ranges. In one range (or stripe), we have martingale solutions, but in the other, we have strict local martingale solutions. The latter correspond to volatility explosion. For some diffusion processes, exact analytical solutions are available. The author discusses the following three important cases: the GARCH diffusion model, the square root model, and the 3/2 model, whereas the solution details are described in Chapter 11. The fundamental transform also has several nice properties and interpretations. For example, it is a generalized Fourier transform, also an (analytical) characteristic function satisfying reflection symmetry and ridge property. Under a certain assumption, it also behaves like a discount bond price. Finally, note that, while the book emphasizes options valuation, the fundamental transform method applies to general financial claims.

When exact solutions are not available, Chapter 3 introduces two series expansion methods to solve the reduced PDE. A deterministic volatility process serves as the benchmark. The expansion, a polynomial of the volatility of volatility (actually a coefficient in the volatility diffusion process), can be made with respect to the price or the implied volatility. In the case of the square root model, both expansion methods provide very accurate approximations to the exact solution, although the latter method performs better when the option is far from the money. This chapter, along with Chapter 2, completely describes the new option valuation method proposed by the author. Applications of the new method to volatility smiles and the term structure of implied volatility are discussed in Chapters 5 and 6. These two chapters, of course, are of immense interest themselves.

Chapter 5 describes the implied volatility as a polynomial of the strike price and applies the fundamental transform to determine the associated coefficients. The volatility expansion method leads to volatility smile at the maturity whenever the price–volatility correlation is negative. The same result can be derived from mixing solution method. This method, the subject of Chapter 4, relies on the property that option prices under stochastic volatility are a weighted sum of option prices under constant volatility. Not surprisingly, there is a correspondence between the mixing weight and the fundamental transform. In terms of computational time (to calculate volatility smiles), the mixing solution method is much more demanding. Once again, the author goes beyond the option pricing and proves that the mixing solution method can be applied to a general financial claim under a general diffusion process with zero price–volatility correlation. Another contribution of Chapter 5 is that the quadratic volatility expansion (i.e., the truncated expansion at the second order) provides very good approximations under various circumstances.

Term structure of implied volatility is referred to the relationship between different implied volatility with different time to expiration holding other parameter values constant. Figs. 6.1 and 6.2 display the relationship for deterministic volatility and square root processes. The remainder of Chapter 6 actually concentrates on the asymptotic volatility (i.e., the implied volatility when the time to expiration approaches infinity). Three solution methods are discussed. It is found that the asymptotic volatility is determined by the volatility diffusion process, and is independent of other parameters. While the mathematical discussions are interesting, I do not find the problem itself of any great value.

If one looks beyond the mathematics, Chapter 7 is perhaps the most important contribution of the book. Frequently, the volatility risk premium is arbitrarily imposed in the literature. The author adopts the Constant Proportional Risk Aversion (CPRA) utility function for a representative agent. Price risk and volatility risk premiums are derived under market equilibrium conditions. The effects of risk aversion on options price are provided for the square root model. The author further argues that the interesting results should hold for a general diffusion process qualitatively.

Chapters 8–10 attribute to the case of infinite volatility. Chapter 9 suggests that volatility explosion (i.e., there is a nonzero probability for the volatility to become infinite within a finite time period) can easily arise within the stochastic volatility model, particularly after the risk adjustment. A generalized pricing formula is proposed. The method essentially adds a correction term that adjusts for volatility explosion. The new solution is related to the change-of-numeraire analysis, a subject discussed in Chapter 8. Finally, Chapter 9 considers the option price when the volatility is very large. Of course, in the real world, we never observe an infinite volatility. Therefore, it cast a doubt on the usefulness of the asymptotic analysis provided in these chapters.

To sum up, this book begins with the premise that a stochastic volatility model is the correct specification. It then compiles many useful mathematical tools to analyze option pricing and the associated properties. For each method, the author provides Mathematica codes to help the readers implement these analyses. Consequently, this is a wonderful book to have for studying and examining option pricing under stochastic volatility processes. It provides a rigorous mathematical journey into the stochastic volatility world. On the other

hand, if someone is looking for intuition or reasoning to engage in stochastic volatility, he will be fairly disappointed. I personally enjoy the book with appreciation of its mathematical depth and rigor, and I would recommend it to academicians and practitioners.

D. Lien

Department of Economics

University of Texas-San Antonio

6900 North Loop, 1604 West, 78249-0631

San Antonio, TX, USA

Tel.: +1-210-4587-312

Fax: +1-210-4585-837

E-mail address: dlien@utsa.edu